Position, distance, displacement

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Distinguish a location, a total path length, and a change in position.

SUCCESS CRITERIA - I can...

1. Label the origin, start, turn and finish.
2. Calculate path distance and signed displacement separately.
3. Explain why position and displacement can differ.

Core: teacher selects from 1-9. **Stretch:** 10 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “position” and “distance” to explain your answer. Sentence starters are optional.

position: location measured from a chosen origin

distance: total length of the path traveled

displacement: change from initial position to final position, with direction

01 Warm-up: bring back an earlier idea

Positions are **15**, **10**, **5 m** at **0**, **2**, **4 s**.
1. **Predict position** at **7 s**.
2. **Give the rate**, something you are treating as true and a future test.

Sentence starter: I predict … . I used … to make this prediction because … .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $\Delta x = x_f - x_i$; distance = sum of path lengths. **Choose an origin** and positive direction. i = initial; f = final.

Worked reasoning: A walker starts at **2 m**, reaches **6 m**, then ends at **4 m**. Distance=**4**+**2**=**6 m**. Displacement=**4**-**2**=**2 m** right.

Your turn: The walker goes from the **2 m** floor mark to the **6 m** mark, then back to **4 m**. On the number line, draw the two legs above the axis and a separate start-to-finish arrow below it. **Label each length**.

Sentence starter: In my work, I used … to show … . I labeled it … .



Position (m); right is positive

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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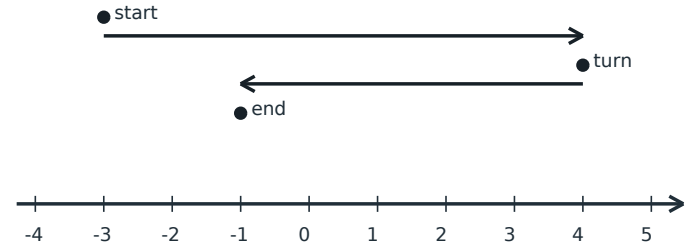
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Track a signed route

A robot moves along an *east*-*west* test track. *East* is positive. It starts at **-3 m**, reaches **+4 m**, then stops at **-1 m**. **Use the route** to **find total distance** and displacement.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Position (m); right is positive

Marzano check - circle: ① ② ③

04 Change the coordinate convention

For the route in **3**, reverse the positive direction while keeping the origin fixed.
1. **Give the new endpoint coordinates** and displacement.
2. What stays unchanged?

Sentence starter: I notice ... . This supports ... because ... .

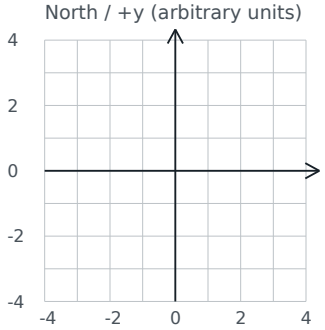
Marzano check - circle: ① ② ③

05 A closed path

A runner completes one **400 m** lap of a circular track, then runs another **100 m**.
1. **Sketch the full trip** and **mark the final point**.
2. Is the displacement magnitude **500 m**?
3. **Explain how the track geometry** determines the start-to-finish distance.

Word help: magnitude = size of a quantity without its direction

Sentence starter: For this answer, I treated ... as true. I found ... . This supports ... because ... .



East / +x (arbitrary units); each square = 1 arbitrary units

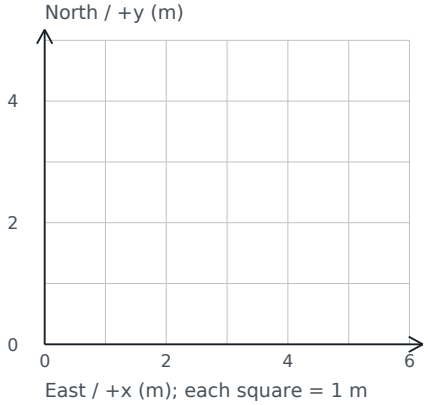
Marzano check - circle: ① ② ③

06 Different paths, same endpoints

A robot must travel from (**0**, **0**) m to (**4**, **3**) m.
1. **Draw two routes** on the grid: one using only *east*/*north* legs and one direct route.
2. **Find each path length** and the displacement magnitude both routes share.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I found ... [unit]. This value means ... in this situation.



East / +x (m); each square = 1 m

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Zero and nonzero

Can displacement be zero while distance is nonzero? Can distance be zero while displacement is nonzero? **Explain why each answer** is correct.

Word help: justify = **support your answer** with a reason or evidence

Sentence starter: For this answer, I treated ... as true. I found ... . This supports ... because ... .

Marzano check - circle: ① ② ③

08 Review: connect an earlier idea

A fictional observing report records **3** images of a moving object but gives no distance or size calibration. **State a supported observation** and **explain why a precise impact** prediction is unsupported.
Separate what was observed from what can be inferred.

Word help: prediction = what a model says may happen next

Sentence starter: I predict ... . I used ... to make this prediction because ... .

Observation	Supported claim	Missing information
___	___	___
___	___	___

Marzano check - circle: ① ② ③

09 Exit check

Start at **+5 m**, walk to **-2 m**, and finish at **+1 m**.
1. **Calculate distance** and displacement.
2. **Explain why neither equals final** position.

Sentence starter: For this answer, I treated ... as true. I found ... . This supports ... because ... .

Marzano check - circle: ① ② ③

10 STRETCH: test the prediction

In one dimension, start at **-2 m** and end at **+3 m** with total distance **11 m**. **Find two possible single-turn positions** and **draw both routes**.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Position (m); right is positive

Marzano check - circle: ① ② ③